

→ [Session 03 페이지 바로가기]

Session 03 – Kubernetes Cluster Provisioning

1. 세션 목적

- Rancher를 통한 멀티 Kubernetes 클러스터 관리 아키텍처를 파악합니다.
- Rancher UI를 통해 신규 클러스터(Downstream)를 생성합니다.
- 이미 구성되어 있는 기존 클러스터를 Rancher 환경으로 가져와 관리합니다.
- 프로비저닝 및 연동된 클러스터의 상태를 점검합니다.

2. 실습 전 확인 사항

- Session 02 완료 및 Rancher Dashboard 정상 접속 가능 여부
- 실습용 VM (`sr-rancher-XX` , `sr-downstream-XX` , `sr-gui-XX`) 정상 부팅 확인
- 각 VM에 SSH 접속 가능한 상태
- 모든 VM 간 원활한 네트워크 통신 상태

3. Hands-on 시나리오



Session 03 실습은 `sr-gui-XX` VM 에서 진행합니다.

VM Password: `SRedu!!!`

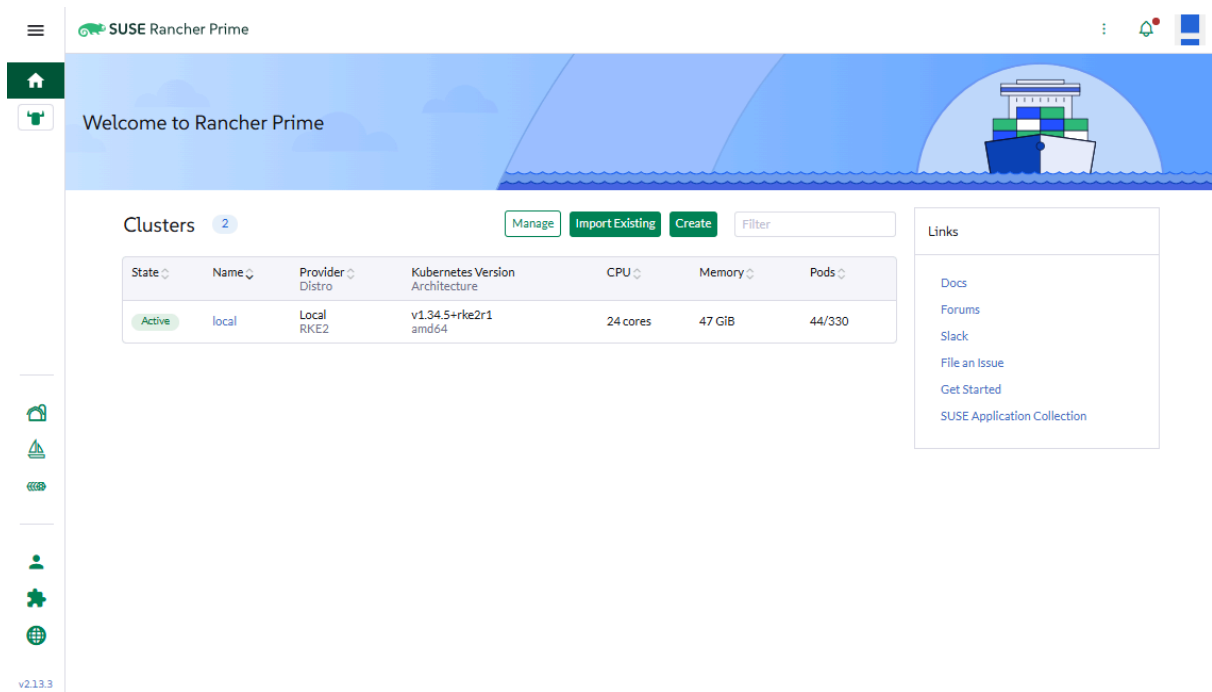
Dashboard admin Password: `RancherEdu2026!`

Step 1. Rancher Dashboard

1-1. Main Dashboard 파악

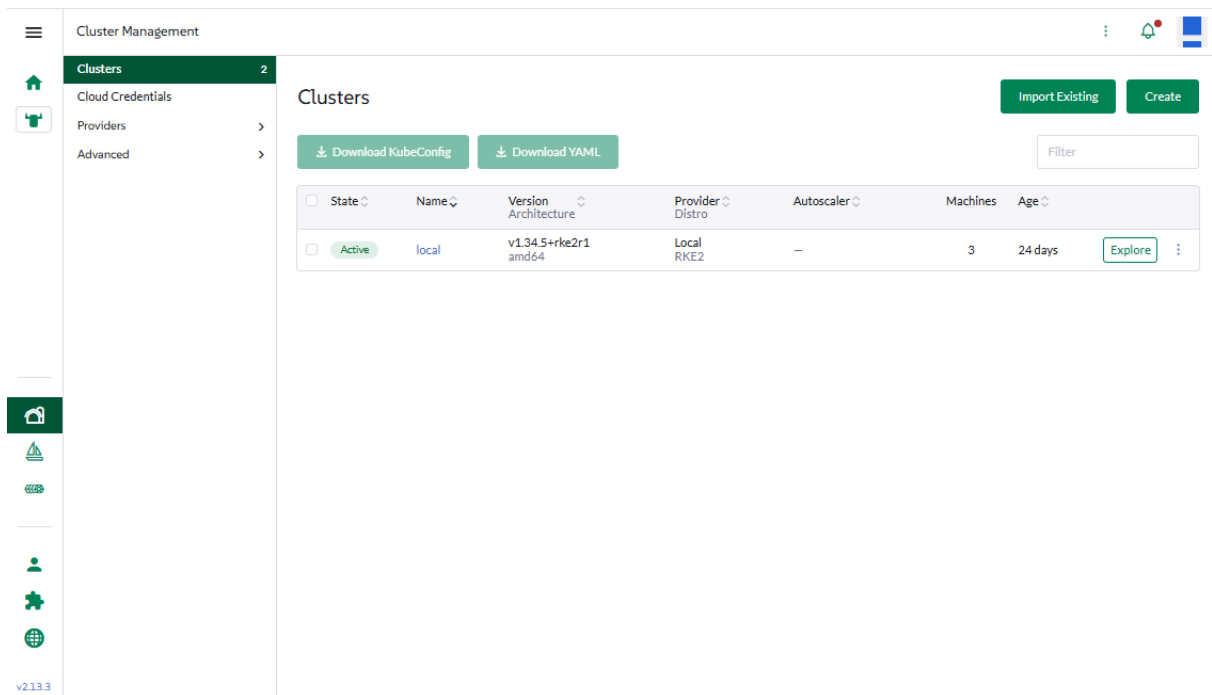
1-1-1. Main

Main Dashboard에서는 등록된 Cluster 목록 확인이 가능합니다.



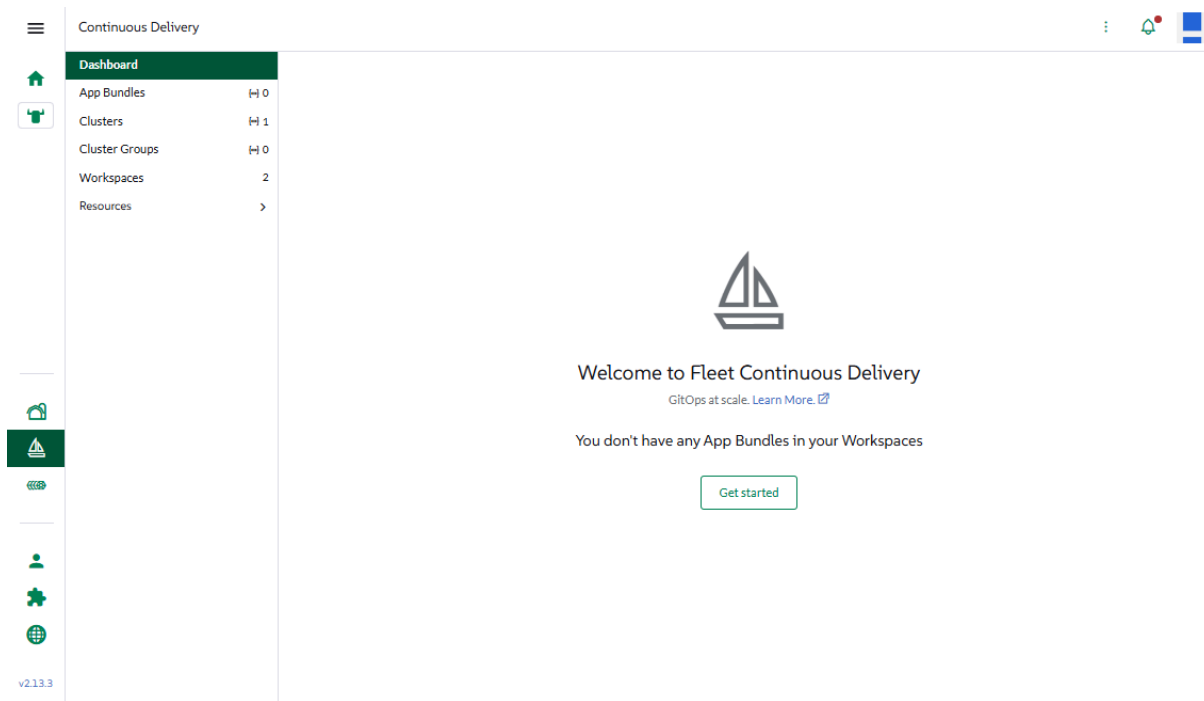
1-1-2. Cluster Management 메뉴

Main > Cluster Management 메뉴에서는 등록된 클러스터 설정 변경, 신규 클러스터 등록이 가능합니다.



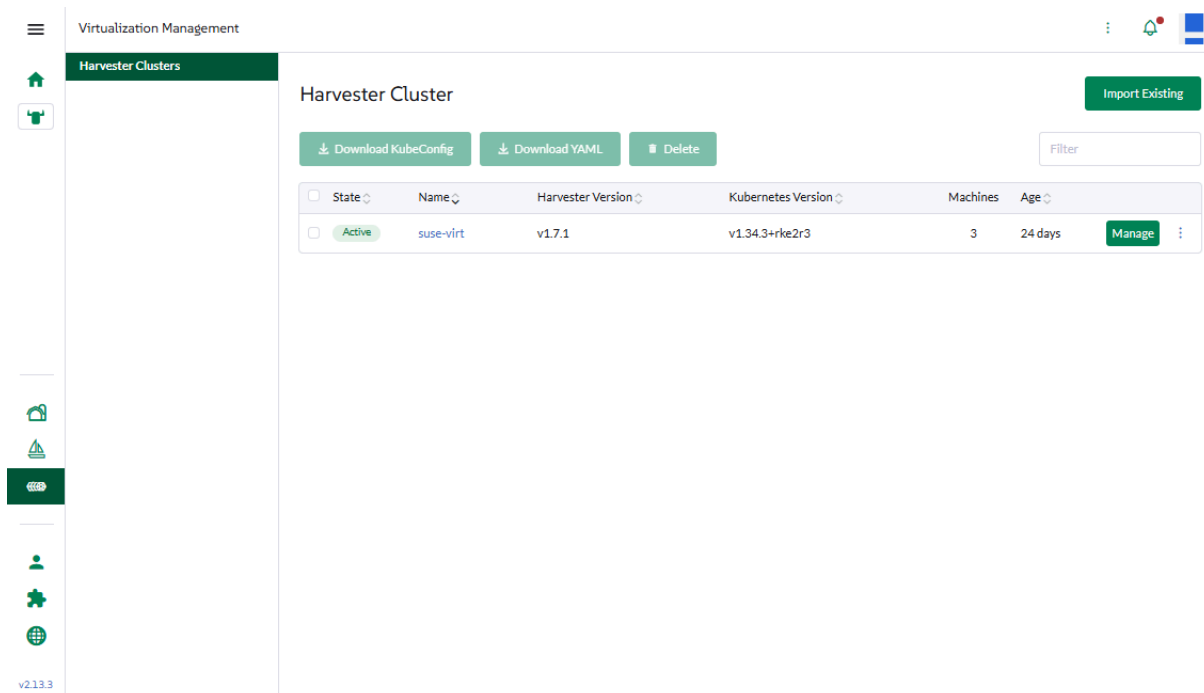
1-1-3. Continuous Delivery 메뉴

Main > Continuous Delivery 메뉴에서는 Fleet을 통한 App 배포를 할 수 있습니다.



1-1-4. Virtualization Management

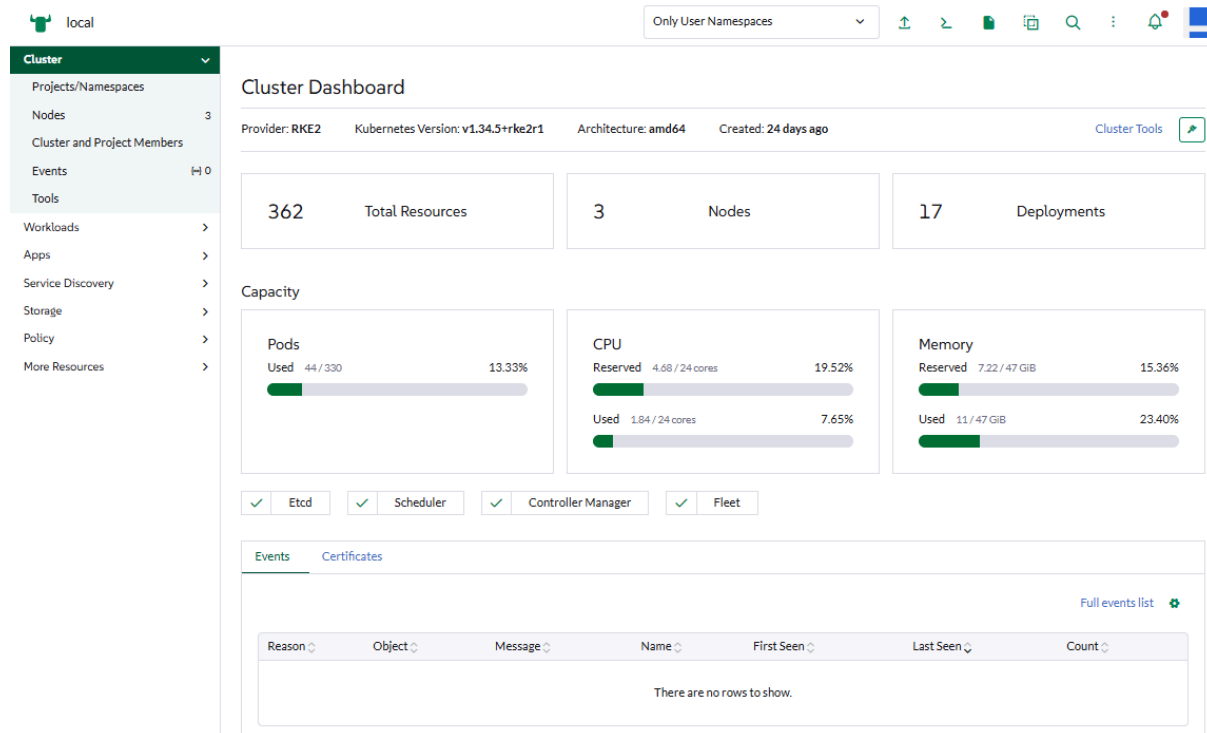
Main > Virtualization Management 메뉴에서는 SUSE-Virtualization 등록을 할 수 있습니다.



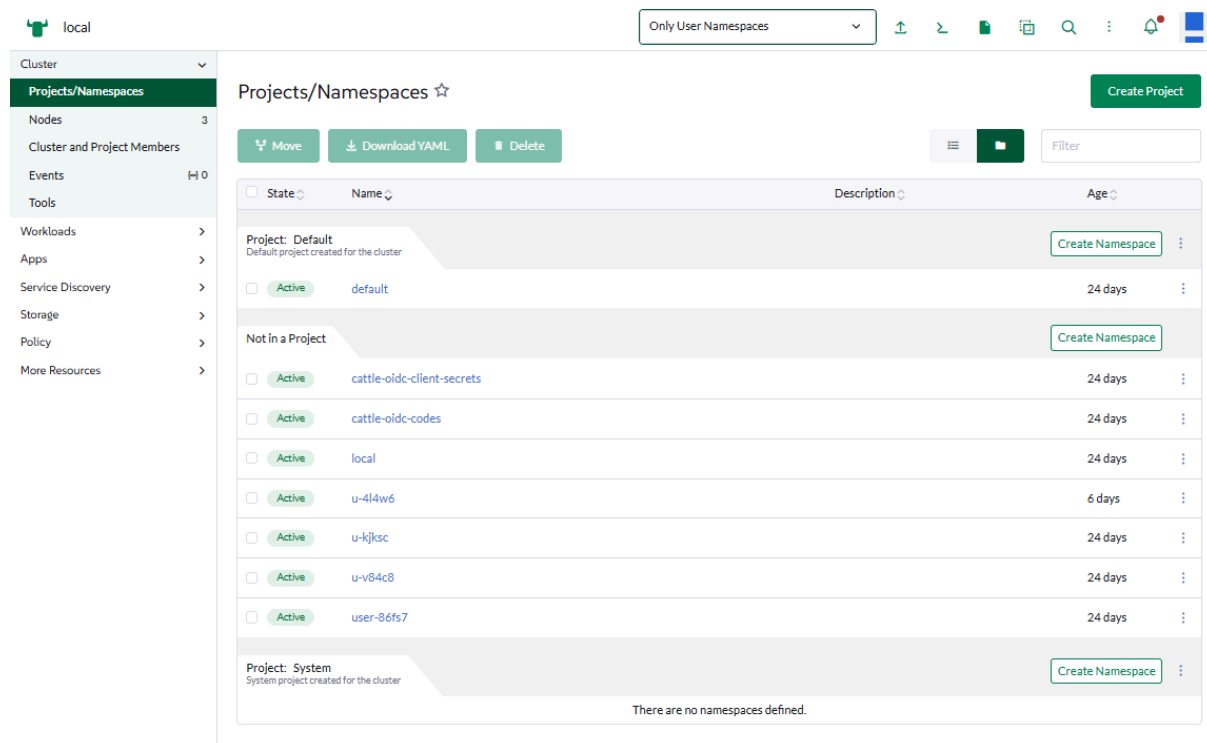
1-2. Cluster Dashboard 파악

1-2-1. Cluster Menu

Cluster (클러스터) 접속 시 확인 가능한 Cluster (클러스터)의 메인 화면입니다.



Cluster > Projects/Namespace 메뉴에서는 Project (프로젝트) 및 Namespace (네임스페이스)를 생성하고 관리하여 Cluster Resource (클러스터 리소스)를 논리적으로 격리할 수 있습니다.



Cluster > Nodes 메뉴에서는 Cluster (클러스터)를 구성하는 Worker (워커) 및 Control Plane Node (컨트롤 플레인 노드)의 상태와 리소스 사용량을 모니터링할 수 있습니다.

The base Kubernetes Node resource represents a virtual or physical machine which hosts deployments. To manage the machine lifecycle, if available, go to Cluster Management.

Nodes ☆

Download YAML Delete Filter

State	Name	Roles	Version	External/Internal IP	OS	CPU	RAM	Pods	Age
Active	rms-01	Control Plane, Etc	v1.34.5+rke2r1	- / 192.168.1.177	Linux	8%	30%	16%	24 days
Active	rms-02	Control Plane, Etc	v1.34.5+rke2r1	- / 192.168.1.178	Linux	5.7%	26%	14%	24 days
Active	rms-03	Control Plane, Etc	v1.34.5+rke2r1	- / 192.168.1.179	Linux	8.2%	31%	13%	24 days

Cluster > Cluster and Project Members 메뉴에서는 Cluster (클러스터) 및 Project (프로젝트) 단위로 사용자 또는 그룹의 접근 권한을 설정할 수 있습니다.

Cluster and Project Members ☆

Delete Add

State	Name	Role	Age
Active	Default Admin (admin)	Cluster Owner	12 hours

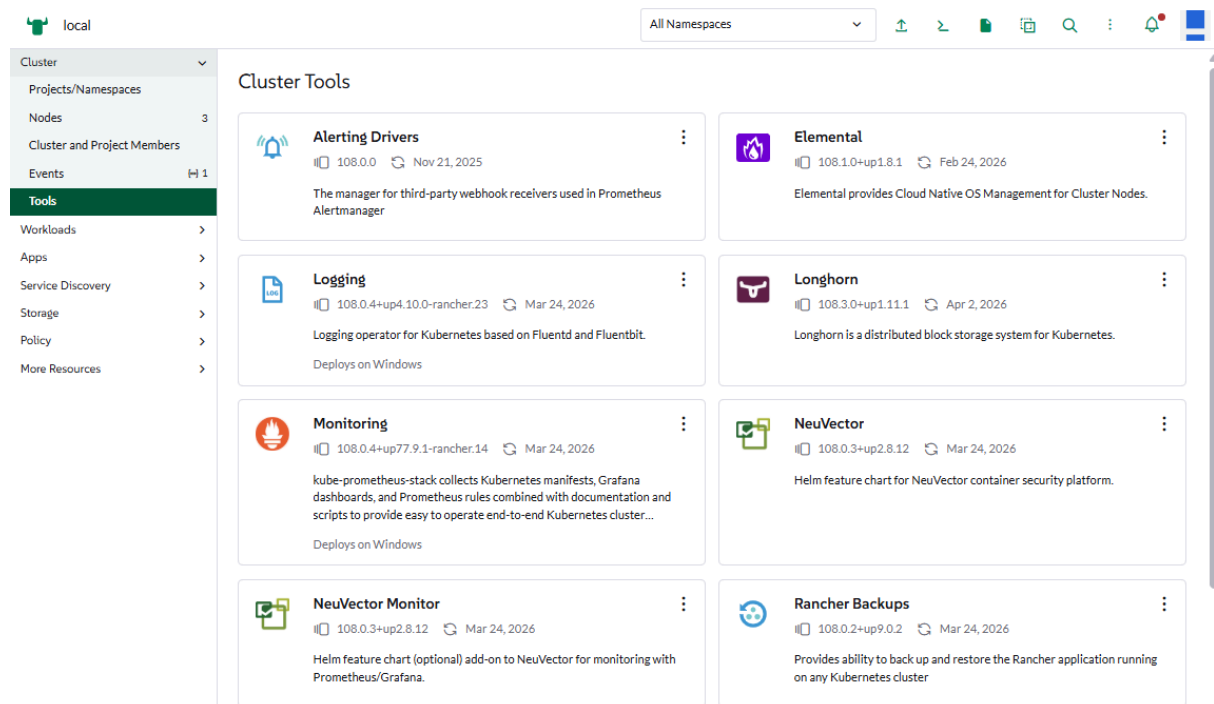
Cluster > Events 메뉴에서는 Cluster (클러스터) 내에서 발생하는 주요 이벤트 및 경고 내역을 확인하여 TroubleShooting (트러블슈팅)에 활용할 수 있습니다.

Events ☆

Download YAML Delete Create from YAML Filter

State	Last Seen	Type	Reason	Object	Subobject	Source	Message	First Seen	Count	Name	Namespace
Active	3m44s	Warning	FailedMount	Pod harvester-cloud-provider-6c74749588-5w59g		kubelet, rms-03	MountVolume.SetUp failed for volume "cloud-config": hostPath type check failed: /var/lib/rancher/rke2/etc/config-files/cloud-provider-config is not a file	78m	45	harvester-cloud-provider-6c74749588-5w59g.18a6c3b84ffa739b	kube-sys

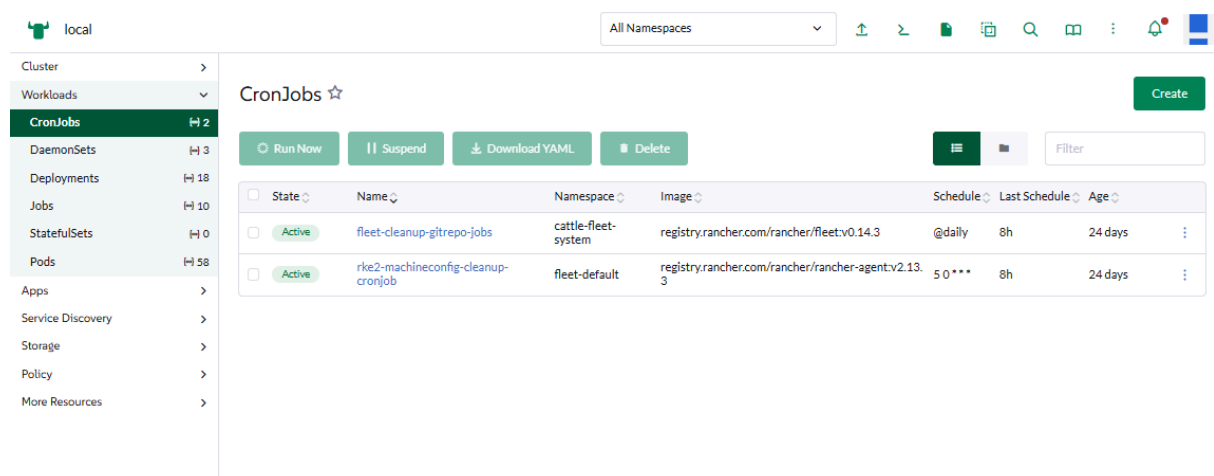
Cluster > Tools 메뉴에서는 Monitoring (모니터링), Logging (로깅), Backup (백업) 등 Cluster (클러스터) 운영에 필요한 확장 도구를 관리할 수 있습니다.



1-2-2. Workloads

Cluster (클러스터) 내에서 실행되는 Application (애플리케이션)의 배포 및 상태를 관리하는 메뉴입니다.

Workloads > CronJobs 메뉴에서는 지정된 시간이나 주기에 따라 주기적으로 실행되는 작업을 관리할 수 있습니다.



Workloads > DaemonSets 메뉴에서는 Cluster (클러스터)의 모든 Node (노드)에서 Pod (파드)가 하나씩 실행되도록 보장하는 Workload (워크로드)를 관리할 수 있습니다.

local

All Namespaces

Cluster >

Workloads >

CronJobs 2

DaemonSets 3

Deployments 18

Jobs 10

StatefulSets 0

Pods 58

Apps >

Service Discovery >

Storage >

Policy >

More Resources >

DaemonSets ☆

Create

Redeploy Download YAML Delete

Filter

State	Name	Namespace	Image	Ready	Current	Desired	Age
Active	kube-vip	kube-system	registry.rancher.com/rancher/mirrored-kube-vip-kube-vip-iptables:v0.9.2	3	3	3	1.9 hours
Active	rke2-canal	kube-system	rancher/hardened-calico:v3.31.3-build20260206	3	3	3	24 days
Active	rke2-ingress-nginx-controller	kube-system	rancher/nginx-ingress-controller:v1.14.3-hardened3	3	3	3	24 days

Workloads > Deployments 메뉴에서는 Application (애플리케이션)의 배포, Scaling (스케일링), Rolling Update (롤링 업데이트)를 관리할 수 있습니다.

Cluster >

Workloads >

CronJobs 2

DaemonSets 2

Deployments 17

Jobs 10

StatefulSets 0

Pods 55

Apps >

Service Discovery >

Storage >

Policy >

More Resources >

Deployments ☆

Create

Redeploy Download YAML Delete

Filter

State	Name	Namespace	Image	Ready	Up To Date	Available	Age
Active	capi-controller-manager	cattle-capi-system	registry.rancher.com/rancher/cluster-api-controller:v1.10.6	1/1	1	1	24 days
Active	cert-manager	cert-manager	quay.io/jetstack/cert-manager-controller:v1.19.1	1/1	1	1	24 days
Active	cert-manager-cainjector	cert-manager	quay.io/jetstack/cert-manager-cainjector:v1.19.1	1/1	1	1	24 days
Active	cert-manager-webhook	cert-manager	quay.io/jetstack/cert-manager-webhook:v1.19.1	1/1	1	1	24 days
Active	fleet-agent	cattle-fleet-local-system	registry.rancher.com/rancher/fleet-agent:v0.14.3	1/1	1	1	24 days
Active	fleet-controller	cattle-fleet-system	registry.rancher.com/rancher/fleet:v0.14.3	1/1	1	1	24 days
Active	gitjob	cattle-fleet-system	registry.rancher.com/rancher/fleet:v0.14.3	1/1	1	1	24 days
Active	helmops	cattle-fleet-system	registry.rancher.com/rancher/fleet:v0.14.3	1/1	1	1	24 days

Workloads > Jobs 메뉴에서는 특정 작업이 완료되면 종료되는 일회성 Workload (워크로드)를 생성하고 관리할 수 있습니다.

Jobs ☆ Create

Download YAML Delete Filter

State	Name	Namespace	Image	Completions	Duration	Age
Active	helm-install-rke2-canal	kube-system	rancher/klipper-helm:v0.9.14-build20260210	1/1	8d	24 days
Active	helm-install-rke2-coredns	kube-system	rancher/klipper-helm:v0.9.14-build20260210	1/1	8d	24 days
Active	helm-install-rke2-ingress-nginx	kube-system	rancher/klipper-helm:v0.9.14-build20260210	1/1	8d	24 days
Active	helm-install-rke2-metrics-server	kube-system	rancher/klipper-helm:v0.9.14-build20260210	1/1	8d	24 days
Active	helm-install-rke2-runtimeclasses	kube-system	rancher/klipper-helm:v0.9.14-build20260210	1/1	8d	24 days
Active	helm-install-rke2-snapshot-controller	kube-system	rancher/klipper-helm:v0.9.14-build20260210	1/1	8d	24 days
Active	helm-install-rke2-snapshot-controller-crd	kube-system	rancher/klipper-helm:v0.9.14-build20260210	1/1	8d	24 days
Active	rke2-machineconfig-cleanup-cronjob-29602085	fleet-default	registry.rancher.com/rancher/rancher-agent:v2.13.3	1/1	2d8h	2.3 days
Active	rke2-machineconfig-cleanup-cronjob-29603525	fleet-default	registry.rancher.com/rancher/rancher-agent:v2.13.3	1/1	32h	1.3 days
Active	rke2-machineconfig-cleanup-cronjob-29604965	fleet-default	registry.rancher.com/rancher/rancher-agent:v2.13.3	1/1	8h	8 hours

Workloads > StatefulSets 메뉴에서는 Database (데이터베이스)와 같이 데이터의 상태 유지가 필요한 Application (애플리케이션)의 배포 및 확장을 관리할 수 있습니다.

StatefulSets ☆ Create

Filter

State	Name	Namespace	Image	Ready	Age
There are no rows to show.					

Workloads > Pods 메뉴에서는 쿠버네티스에서 실행되는 가장 작은 배포 단위인 Pod (파드)를 확인할 수 있습니다.

Pods ☆

Download YAML Delete

State	Name	Namespace	Image	Ready	Restarts	IP	Node	Age
Running	capi-controller-manager-78558fffd-s54hb	cattle-capi-system	registry.rancher.com/rancher/cluster-api-controller:v1.10.6	1/1	0	10.42.0.18	rms-01	24 days
Running	cert-manager-55c5fdb5f5-ww5fl	cert-manager	quay.io/jetstack/cert-manager-controller:v1.19.1	1/1	0	10.42.2.3	rms-03	24 days
Running	cert-manager-cainjector-5dc4bf4cdf-qkg89	cert-manager	quay.io/jetstack/cert-manager-cainjector:v1.19.1	1/1	0	10.42.2.4	rms-03	24 days
Running	cert-manager-webhook-6ff7dcb868-scp7t	cert-manager	quay.io/jetstack/cert-manager-webhook:v1.19.1	1/1	0	10.42.1.4	rms-02	24 days
Running	cloud-controller-manager-rms-01	kube-system	rancher/rke2-cloud-provider:v1.34.4-0.20260211145917-c6017918a65-c-build20260211	1/1	1 (22h ago)	192.168.1.177	rms-01	24 days
Running	cloud-controller-manager-rms-02	kube-system	rancher/rke2-cloud-provider:v1.34.4-0.20260211145917-c6017918a65-c-build20260211	1/1	0	192.168.1.178	rms-02	24 days
Running	cloud-controller-manager-rms-03	kube-system	rancher/rke2-cloud-provider:v1.34.4-0.20260211145917-c6017918a65-c-build20260211	1/1	1 (16s ago)	192.168.1.179	rms-03	24 days
Running	etcd-rms-01	kube-system	rancher/hardened-etcd:v3.6.7-k3s1-build20260227	1/1	0	192.168.1.177	rms-01	24 days
Running	etcd-rms-02	kube-system	rancher/hardened-etcd:v3.6.7-k3s1-build20260227	1/1	0	192.168.1.178	rms-02	24 days
Running	etcd-rms-03	kube-system	rancher/hardened-etcd:v3.6.7-k3s1-build20260227	1/1	0	192.168.1.179	rms-03	24 days

1-2-3. Apps

Helm Chart (헬름 차트) 기반의 Application Catalog (애플리케이션 카탈로그)를 관리하고 배포하는 메뉴입니다.

Apps > Charts 메뉴에서는 Rancher Catalog 또는 Helm Repository에서 제공하는 애플리케이션 목록을 조회하고 설치할 수 있습니다.

Charts

Refresh all repositories

Search the catalogue...

Repository

- ☐ Rancher Prime
- ☐ Partners
- ☐ rancher
- ☐ RKE2
- [+ Add new](#)

Category

- ☐ Applications
- ☐ Database
- ☐ Infrastructure
- ☐ Monitoring
- ☐ Networking
- ☐ PaaS
- ☐ Security
- ☐ Storage

139 charts in total

Sort by: Recommended

DxOperator

1.0.0 Mar 18, 2026

FEAT DH2i operator for Kubernetes

Partners Linux Only

JFrog Artifactory HA

107.133.18 Apr 16, 2026

FEAT Universal Repository Manager supporting all major packaging formats, build tools and CI servers.

Partners Linux Only

Kasten

8.5.601 Apr 8, 2026

FEAT Kasten's K10 Data Management Platform

Partners Linux Only

CloudCasa

3.4.7 Mar 20, 2026

FEAT CloudCasa backup, DR, and migration service agent for Kubernetes

Partners Linux Only

Apps > Installed Apps 메뉴에서는 현재 Cluster (클러스터)에 설치된 Application (애플리케이션)의 버전을 확인하거나 업그레이드, 삭제를 수행할 수 있습니다.

local

All Namespaces

Cluster >

Workloads >

Apps >

Charts

Installed Apps 16

Repositories 4

Recent Operations 1

Service Discovery >

Storage >

Policy >

More Resources >

An installed application is a Helm 3 chart that was installed either via our charts or through the Helm CLI.

Installed Apps ☆

Download YAML Delete

State	Name	Namespace	Chart	Upgradable	Resources	Age
Deployed	cert-manager	cert-manager	cert-manager:1.19.1		42	24 days
Deployed	fleet	cattle-fleet-system	fleet:108.0.3+up0.14.3		24	24 days
Deployed	fleet-agent-local	cattle-fleet-local-system	fleet-agent-local:0.0.0+s-dfbc903	Managed	6	24 days
Deployed	fleet-crd	cattle-fleet-system	fleet-crd:108.0.3+up0.14.3		13	24 days
Deployed	harvester	cattle-ui-plugin-system	harvester:1.7.1		1	24 days
Deployed	rancher	cattle-system	rancher:2.13.3		8	24 days
Deployed	rancher-turtles	cattle-turtles-system	rancher-turtles:108.0.4+up0.25.4	Managed	13	24 days
Deployed	rancher-webhook	cattle-system	rancher-webhook:108.0.3+up0.9.3	Managed	7	24 days
Deployed	rke2-canal	kube-system	rke2-canal:3.31.3-build2026020600	Managed	31	24 days
Deployed	rke2-coredns	kube-system	rke2-coredns:1.45.201	Managed	11	24 days

Apps > Repositories 메뉴에서는 Application (애플리케이션) 설치에 사용할 외부 Helm Chart (헬름 차트) 저장소를 추가하고 관리할 수 있습니다.

local

All Namespaces

Cluster >

Workloads >

Apps >

Charts

Installed Apps 16

Repositories 4

Recent Operations 1

Service Discovery >

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More Resources >

A chart repository is a Helm repository or Rancher Prime git based application catalog. It provides the list of available charts in the cluster. Cluster Templates are deployed via Helm charts.

Repositories ☆

Create

Refresh Disable Download YAML Delete

State	Name	Type	URL	Branch	Age
Active	Partners	git	https://git.rancher.io/partner-charts	main	24 days
Active	rancher	git	https://github.com/rancher/ui-plugin-charts	main	24 days
Active	Rancher Prime	git	https://git.rancher.io/charts	release-v2.13	24 days
Active	RKE2	git	https://git.rancher.io/rke2-charts	main	24 days

Apps > Recent Operations 메뉴에서는 Application (애플리케이션)의 설치, 업데이트, 삭제 등 최근에 수행된 작업의 성공 여부를 확인할 수 있습니다.

local

All Namespaces

Recent Operations ☆

Download YAML Delete

State	Name	Namespace	Action	Release Namespace	Release Name	Age
Active	helm-operation-8lj7h	kube-system	install	kube-system	harvester-cloud-provider	33 secs

1-2-4. Service Discovery

Service Discovery > HorizontalPodAutoscalers 메뉴에서는 CPU, Memory (메모리) 등의 Resource (리소스) 사용량을 기반으로 Pod (파드)의 개수를 자동으로 조절하는 설정을 관리할 수 있습니다.

local

All Namespaces

HorizontalPodAutoscalers ☆

Create

State	Name	Namespace	Workload	Minimum Replicas	Maximum Replicas	Current Replicas	Age
There are no rows to show.							

Service Discovery > Ingresses 메뉴에서는 외부에서 Cluster (클러스터) 내부의 Service (서비스)로 접근할 수 있도록 HTTP/HTTPS Routing (라우팅) 규칙을 설정할 수 있습니다.

The screenshot shows the Rancher UI interface. On the left, the 'Service Discovery' menu is expanded, and 'Ingresses' is selected. The main content area displays the 'Ingresses' page for the 'local' cluster. It includes a 'Create' button and a table with the following data:

State	Name	Namespace	Target	Default	Ingress Class	Age
Active	rancher	cattle-system	https://192.168.1.177.nip.io/		rancher	24 days

Service Discovery > Services 메뉴에서는 Pod (파드)가 재생성되더라도 내부 네트워크 통신이 끊기지 않도록 고정된 연결 통로를 제공합니다.

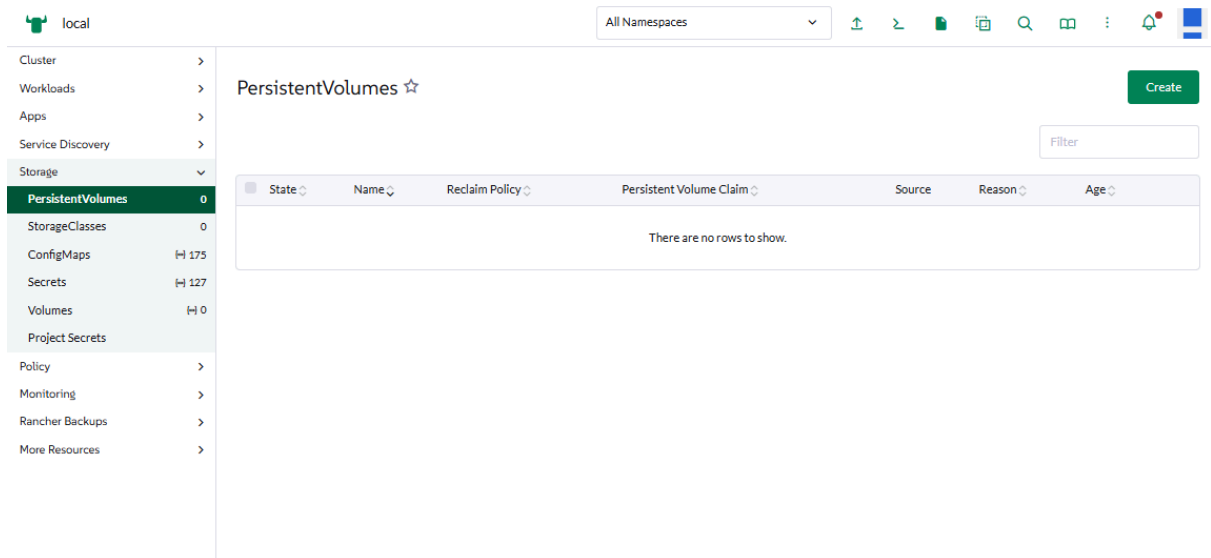
The screenshot shows the Rancher UI interface with the 'Services' page selected. The main content area displays a table with the following data:

State	Name	Namespace	Target	Selector	Type	Age
Active	alertmanager-operated	cattle-monitoring-system	9093 http-web/TCP 9094 mesh-tcp/TCP 9094 mesh-udp/UDP	app.kubernetes.io/name=alertmanager	Cluster IP (Headless)	12 hours
Active	capi-webhook-service	cattle-capi-system	10.43.161.28:443 webhook-server/TCP	cluster.x-k8s.io/provider=cluster-api	Cluster IP	24 days
Active	cert-manager	cert-manager	10.43.30.166:9402 http-metrics/TCP	app.kubernetes.io/component=controller app.kubernetes.io/instance=cert-manager app.kubernetes.io/name=cert-manager	Cluster IP	24 days
Active	cert-manager-cainjector	cert-manager	10.43.220.220:9402 http-metrics/TCP	app.kubernetes.io/component=cainjector app.kubernetes.io/instance=cert-manager app.kubernetes.io/name=cainjector	Cluster IP	24 days
Active	cert-manager-webhook	cert-manager	https http/TCP 10.43.30.106:9402 http-metrics/TCP	app.kubernetes.io/component=webhook app.kubernetes.io/instance=cert-manager app.kubernetes.io/name=webhook	Cluster IP	24 days

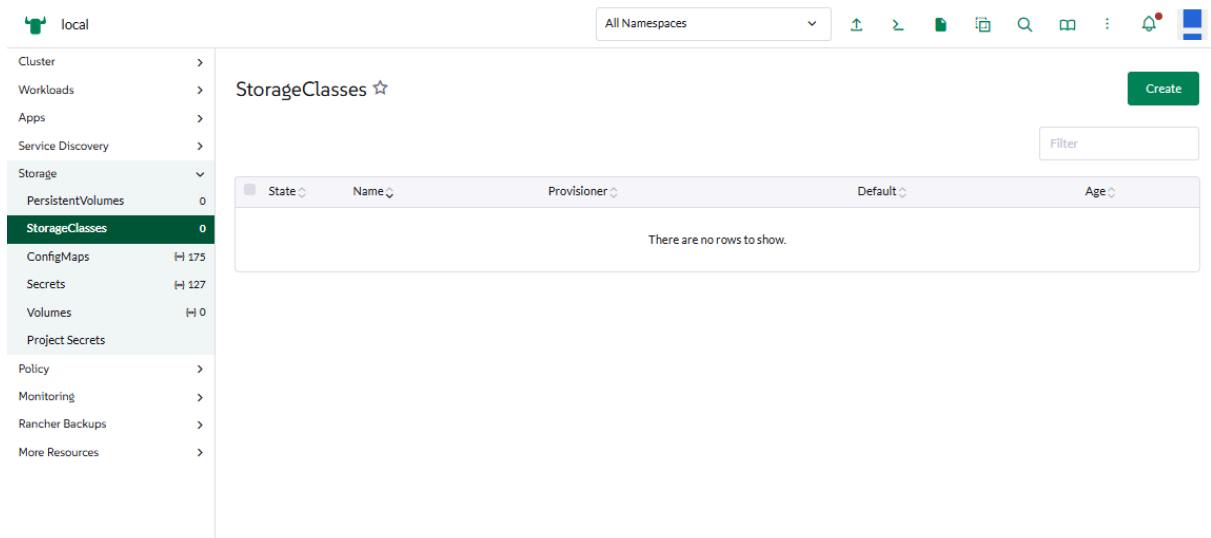
1-2-5. Storage

Application (애플리케이션)이 사용할 영구 저장소 및 설정 데이터, 보안 정보를 관리하는 메뉴입니다.

Storage > PersistentVolumes 메뉴에서는 관리자가 생성한 스토리지 볼륨(PV)의 목록과 상태를 확인할 수 있습니다.



Storage > StorageClasses 메뉴에서는 동적 Volume (볼륨) 생성을 위한 StorageClass (스토리지 클래스) 목록과 정책을 확인할 수 있습니다.



Storage > ConfigMaps 메뉴에서는 Container Image (컨테이너 이미지) 변경 없이 Application (애플리케이션) 설정 데이터를 키-값 쌍으로 저장하고 주입할 수 있습니다.

local

All Namespaces

Cluster > Workloads > Apps > Service Discovery > Storage > PersistentVolumes 0 > StorageClasses 0 > **ConfigMaps 175** > Secrets 127 > Volumes 0 > Project Secrets > Policy > Monitoring > Rancher Backups > More Resources >

ConfigMaps ☆

Download YAML Delete

Create

Filter

Name	Namespace	Data	Age
ad-guid-migration	cattle-system	ad-guid-migration-status, statusLastUpdated	24 days
admincreated	cattle-system	CleanupOrphanBindingsDone, DedupeBindingsDone	24 days
capi-additional-rbac-roles	cattle-capi-system	manifests	24 days
chart-content-rke2-canal	kube-system	rke2-canal.tgz.base64	25 days
chart-content-rke2-coredns	kube-system	rke2-coredns.tgz.base64	25 days
chart-content-rke2-ingress-nginx	kube-system	rke2-ingress-nginx.tgz.base64	25 days
chart-content-rke2-metrics-server	kube-system	rke2-metrics-server.tgz.base64	25 days
chart-content-rke2-runtimeclasses	kube-system	rke2-runtimeclasses.tgz.base64	25 days
chart-content-rke2-snapshot-controller	kube-system	rke2-snapshot-controller.tgz.base64	25 days
chart-content-rke2-snapshot-controller-crd	kube-system	rke2-snapshot-controller-crd.tgz.base64	25 days
cluster-api-operator-resources-cleanup-script	cattle-turtles-system	cleanup.sh	24 days

Storage > Secrets 메뉴에서는 비밀번호, 토큰, 인증서 등 민감한 정보를 안전하게 저장하고 컨테이너에 제공할 수 있습니다.

local

All Namespaces

Cluster > Workloads > Apps > Service Discovery > Storage > PersistentVolumes 0 > StorageClasses 0 > ConfigMaps 175 > **Secrets 127** > Volumes 0 > Project Secrets > Policy > Monitoring > Rancher Backups > More Resources >

Secrets ☆

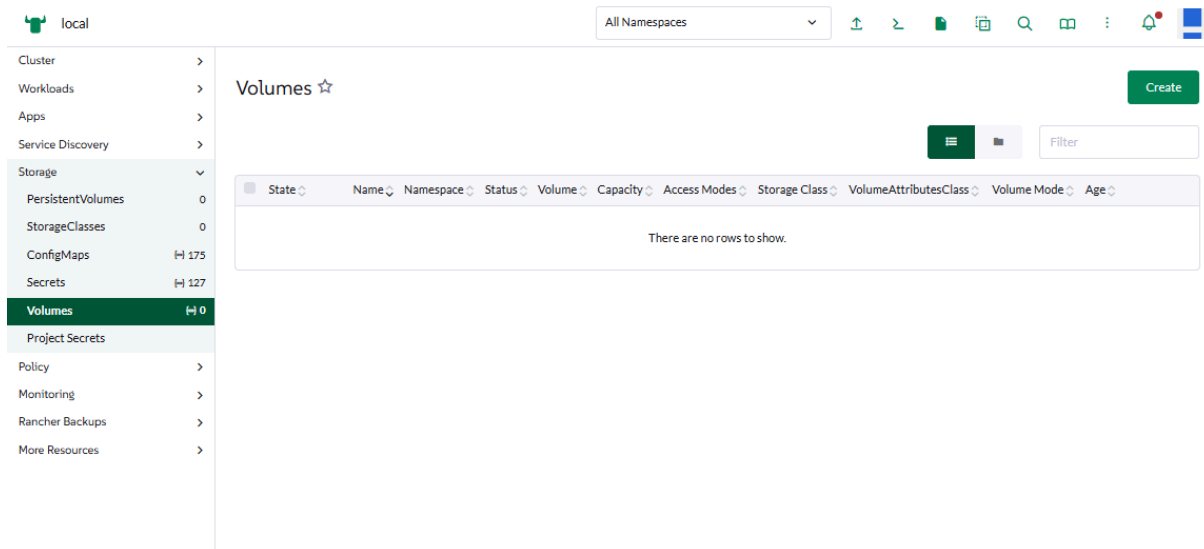
Download YAML Delete

Create

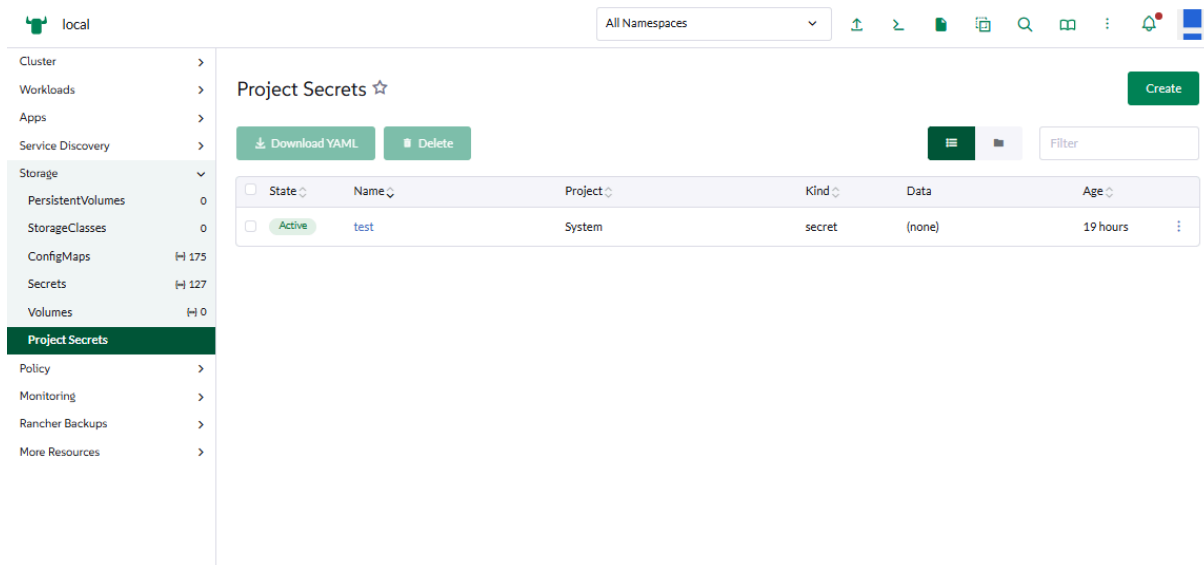
Filter

State	Name	Namespace	Kind	Data	Created by Project Secret	Project Secret	Age
Active	alertmanager-rancher-monitoring-alertmanager	cattle-monitoring-system	Opaque	alertmanager.yaml, rancher_defaults.tpl	—	—	12 hours
Active	alertmanager-rancher-monitoring-alertmanager-cluster-tls-config	cattle-monitoring-system	Opaque	cluster-tls-config.yaml	—	—	12 hours
Active	alertmanager-rancher-monitoring-alertmanager-generated	cattle-monitoring-system	Opaque	alertmanager.yaml.gz, rancher_defaults.tpl	—	—	12 hours
Active	alertmanager-rancher-monitoring-alertmanager-tls-assets-0	cattle-monitoring-system	Opaque	(none)	—	—	12 hours
Active	alertmanager-rancher-monitoring-alertmanager-web-config	cattle-monitoring-system	Opaque	web-config.yaml	—	—	12 hours
Active	bootstrap-secret	cattle-system	Opaque	bootstrapPassword	—	—	24 days
Active	c-zw74q-kubeconfig	fleet-default	Opaque	token, value	—	—	24 days
Active	capi-env-variables	cattle-capi-system	Opaque	CLUSTER_TOPOLOGY, EXP_CLUSTER_RESOURCE_SET, EXP_MACHINE_POOL	—	—	24 days

Storage > Volumes 메뉴에서는 Pod (파드)가 사용 중인 PVC 목록과 연결 상태를 확인할 수 있습니다.



Storage > Project Secrets 메뉴에서는 특정 Project (프로젝트) 안에 있는 여러 Namespace (네임스페이스) 들이 함께 공유해서 사용할 수 있는 보안 정보들을 한 곳에서 관리할 수 있습니다.



Step 2. Downstream Cluster 생성

2-1. Custom Cluster Create

2-1-1. sr-downstream-XX VM 생성 확인

sr-downstream-XX VM의 IP Address를 저장합니다.

Virtual Machines

Create

State	Name	CPU	Memory	IP Address	Node	Age
Running	sr-import-99	4	8 Gi	192.168.2.92	suse-virt02	6 days
Running	sr-downstream-99	4	8 Gi	192.168.2.93	suse-virt01	6 days
Running	sr-gui-99	2	4 Gi	192.168.2.91	suse-virt02	6 days
Running	sr-rancher-99	4	8 Gi	192.168.2.90	suse-virt02	6 days

2-1-2. WebVNC 콘솔 접속

`sr-downstream-xx` VM의 **Console** 버튼을 클릭한 뒤 **Open in WebVNC**를 선택합니다.

Virtual Machines

Create

State	Name	CPU	Memory	IP Address	Node	Age
Running	harvester-test-hav-cluster-gn8d4-cs7g7	4	8 Gi	192.168.2.104	suse-virt02	28 mins
Running	sr-import-99	4	8 Gi	192.168.2.92	suse-virt02	6 days
Running	sr-downstream-99	4	8 Gi	192.168.2.93	suse-virt01	6 days
Running	sr-gui-99	2	4 Gi	192.168.2.91	suse-virt02	6 days
Running	sr-rancher-99	4	8 Gi	192.168.2.90	suse-virt02	7 days

팁: 이후 모든 명령어는 이 WebVNC 콘솔 화면에서 입력합니다.

2-1-3. 시스템 업데이트

시스템 패키지를 최신 상태로 업데이트합니다.

```
zypper update
```

! 시스템 패키지 업데이트는 5분 이상 소요될 수 있습니다.

2-1-4. 필수 패키지 설치

RKE2 및 쿠버네티스 클러스터 운영에 필요한 필수 패키지(`iptables`)를 설치합니다.


```
zypper install -y iptables
```

```
iptables libip6tc2 libnetfilter_conntrack3 libnfnetlink0 libnftnl11 xtables-plugins
6 new packages to install.
Package download size: 620.6 KiB
Package install size change:
    2.3 MiB required by packages that will be installed
    2.3 MiB - 0 B released by packages that will be removed
Backend: classic_rpmtrans
Continue? [y/n/v/...? shows all options] (y): y
Retrieving: libip6tc2-1.8.7-1.1.x86_64 (SLE-Module-Basesystem15-SP7-Pool) (1/6), 24.4 KiB
Retrieving: libip6tc2-1.8.7-1.1.x86_64.rpm .....[done]
Retrieving: libnfnetlink0-1.0.1-2.11.x86_64 (SLE-Module-Basesystem15-SP7-Pool) (2/6), 27.7 KiB
Retrieving: libnfnetlink0-1.0.1-2.11.x86_64.rpm .....[done]
Retrieving: libnftnl11-1.2.0-150400.1.6.x86_64 (SLE-Module-Basesystem15-SP7-Pool) (3/6), 67.3 KiB
Retrieving: libnftnl11-1.2.0-150400.1.6.x86_64.rpm .....[done]
Retrieving: libnetfilter_conntrack3-1.0.7-1.38.x86_64 (SLE-Module-Basesystem15-SP7-Pool) (4/6), 47.7 KiB
Retrieving: libnetfilter_conntrack3-1.0.7-1.38.x86_64.rpm .....[done]
Retrieving: xtables-plugins-1.8.7-1.1.x86_64 (SLE-Module-Basesystem15-SP7-Pool) (5/6), 243.4 KiB
Retrieving: xtables-plugins-1.8.7-1.1.x86_64.rpm .....[done]
Retrieving: iptables-1.8.7-1.1.x86_64 (SLE-Module-Basesystem15-SP7-Pool) (6/6), 210.2 KiB
Retrieving: iptables-1.8.7-1.1.x86_64.rpm .....[done]
Checking for file conflicts: .....[done]
(1/6) Installing: libip6tc2-1.8.7-1.1.x86_64 .....[done]
(2/6) Installing: libnfnetlink0-1.0.1-2.11.x86_64 .....[done]
(3/6) Installing: libnftnl11-1.2.0-150400.1.6.x86_64 .....[done]
(4/6) Installing: libnetfilter_conntrack3-1.0.7-1.38.x86_64 .....[done]
(5/6) Installing: xtables-plugins-1.8.7-1.1.x86_64 .....[done]
update-alternatives: using /usr/sbin/xtables-legacy-multi to provide /usr/sbin/iptables (iptables) in auto mode
(6/6) Installing: iptables-1.8.7-1.1.x86_64 .....[done]
```

✓ **체크포인트:** `iptables --version` 명령으로 iptables가 정상 설치되었는지 확인합니다.

```
iptables --version
```

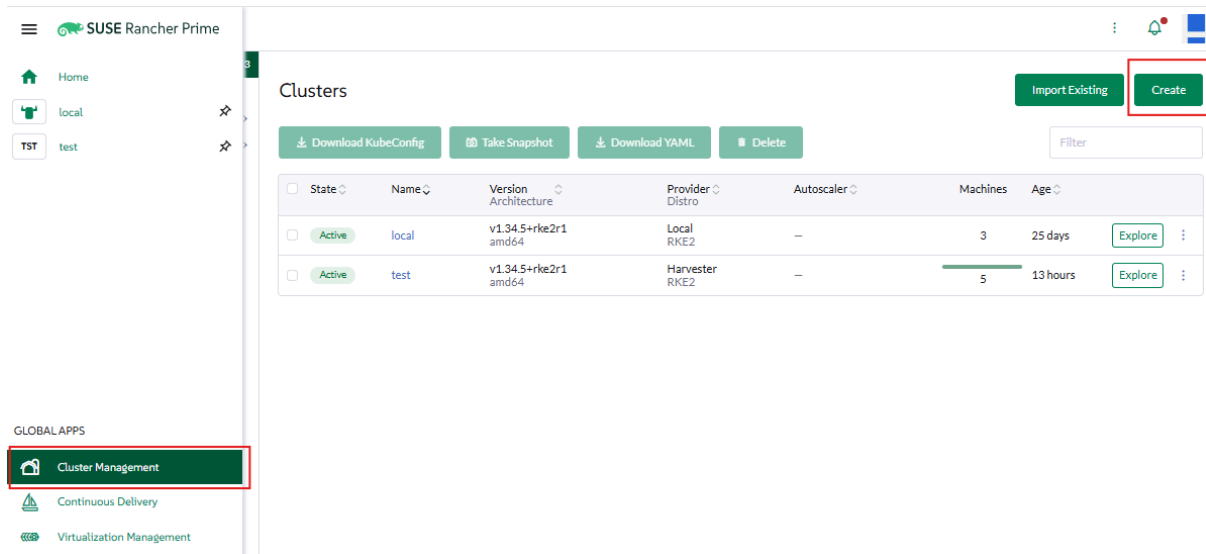
2-1-5. Downstream cluster create

SUSE Rancher Prime에서는 클러스터를 "직접 생성(Provision)"하는 방식과 "기존 클러스터를 가져오는(Import)" 방식이 있습니다. 이 섹션에서는 직접 생성 방식 중 가장 유연한 **Custom 방식**을 다룹니다.

Main > Cluster Management 메뉴에 접속 후 **Create** 버튼을 클릭합니다.

SUSE Rancher Prime은 다양한 인프라 환경에서 Kubernetes 클러스터를 생성하고 통합 관리할 수 있습니다.

Public Cloud(Amazon EKS, Azure AKS, Google GKE)뿐만 아니라, On-Premise 및 Private Cloud 환경(VMware vSphere, Harvester 등)에서도 클러스터를 구성할 수 있으며, 이를 하나의 플랫폼에서 통합 관리함으로써 Hybrid Cloud 환경을 유연하게 구축할 수 있습니다.



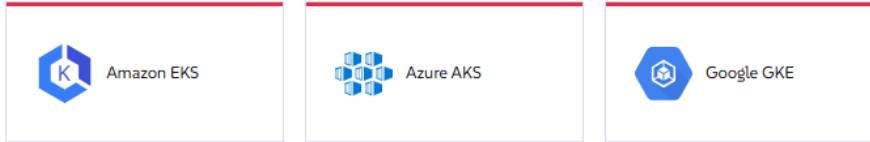
Custom 을 선택합니다.

Custom 옵션은 특정 CSP(Cloud Service Provider)에 종속되지 않고, 사용자가 직접 노드 구성 및 네트워크 설정을 제어할 수 있는 방식입니다.

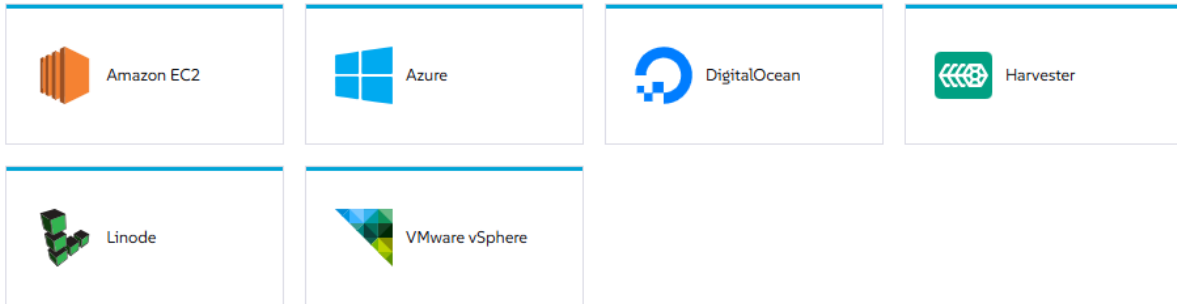
본 실습에서는 사내 인프라(On-Premise VM 환경)를 활용하므로, 클러스터 노드를 직접 등록하고 구성할 수 있는 **Custom** 방식을 사용합니다.

Cluster: Create

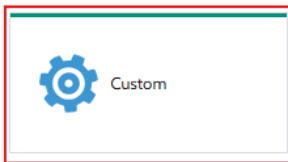
Create a cluster in a hosted Kubernetes provider



Provision new nodes and create a cluster using RKE2/K3s



Use existing nodes and create a cluster using RKE2/K3s



Cluster Name 을 입력하고 **Kubernetes Version** 을 선택한 후, **Create** 버튼을 클릭합니다.

- **Cluster Name:** `custom-test`
- **Kubernetes Version:** `v1.33.10+rke2r3`

Cluster: Create Custom

Cluster Name *

custom-test

Cluster Appearance

CMT

Customize

Cluster Description

Any text you want that better describes this cluster

Cluster Configuration

Basics

Member Roles

Add-on: Calico

Additional Manifest

Agent Environment Vars

Cluster Agent

etcd

Fleet Agent

Labels & Annotations

Networking

Registries

Upgrade Strategy

Advanced

Basics

On Kubernetes 1.30 or greater, the Azure Cloud Provider has been removed. See [the documentation](#) for more information.

Kubernetes Version

v1.33.10+rke2r3

Cloud Provider

Default - RKE2 Embedded

☐ Show deprecated Kubernetes patch versions

Container Network

calico

Security

Compliance Profile

(None)

Pod Security Admission Configuration Template

Default - RKE2 Embedded

Cancel

Edit as YAML

Create

Step 1에 **etcd**, **Control Plane**, **Worker** 를 전부 체크 후 Step 2에 **Insecure** 를 체크합니다.

아래 화면은 Rancher에서 커스텀 클러스터(**custom-test**)에 노드를 등록하는 Registration 탭 화면입니다. 노드 역할(Node Role) 선택과 등록 명령어 확인 두 단계로 구성되어 있습니다.

Cluster: custom-test Updating Explore Show Configuration

Namespace fleet-default Labels and Annotations 0
 Age 2 secs There are no labels and annotations configured on this resource.
 Provisioner RKE2 [Show Configuration](#)

Machines (0) Provisioning Log **Registration** Snapshots (0) Autoscaler (0) Conditions (16) Recent Events (0) Related Resources (10)

Step 1

Node Role
 Choose what roles the node will have in the cluster. The cluster needs to have at least one node with each role.

☒ etcd ☒ Control Plane ☒ Worker

[Show Advanced](#)

Step 2

Registration Command
 Run this command on each of the existing Linux machines you want to register.

```
curl --insecure -fL https://192.168.1.177.nip.io/system-agent-install.sh | sudo sh -s --server https://192.168.1.177.nip.io --label 'cattle.io/os=linux' --token 2jq5dshk257tdjff7cgmgcg4267mlf61xrbnfqscn88qgj47mhp9r1 --ca-checksum b7dd02f7377e226897967b5fd1cd06616a33e7b0a58b71b2301cf01f85937351 --etcd --controlplane --worker
```

☒ Insecure: Select this to skip TLS verification if your server has a self-signed certificate.

Run this command in PowerShell on each of the existing Windows machines you want to register.

The cluster must be up and running with Linux etcd, control plane, and worker nodes before the registration command for adding Windows workers will display.

Windows nodes can only be workers.

etcd는 클러스터의 상태 정보를 저장하는 분산 key-value 데이터베이스 역할을 합니다. Control Plane은 클러스터 전체를 제어하는 마스터 역할이며, Worker는 실제 워크로드 (Pod)가 실행되는 노드 역할입니다. 본 실습에서는 단일 VM 에 세 가지 역할(etcd, Control Plane, Worker)을 모두 부여하여 All-in-One 구성으로 진행합니다.

2-1-1에서 확인한 IP 주소를 사용하여 SSH로 **sr-downstream-XX** VM VNC Console 에 접속합니다.

접속 후, Step 2에서 확인한 **Registration Command** 를 복사하여 터미널에서 실행합니다.

```

sr-downstream-99:~ # curl --insecure -fL https://192.168.1.177.nip.io/system-agent-install.sh | sudo sh -s - --server https://192.168.1.177.nip.io --label 'cattle.io/os=linux' --token 2jq5dshk257tdjf7cgmkgc4267mlf6lxrnfqscn88qgj47mhp9rl --ca-checksum b7dd02f7377e226897967b5fd1cd06616a33e7b0a58b71b2301cf01f8593735
1 --etcd --controlplane --worker
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 33679  0 33679  0  0  617k    0 --:--:-- --:--:-- --:--:-- 644k
[INFO] Label: cattle.io/os=linux
[INFO] Role requested: etcd
[INFO] Role requested: controlplane
[INFO] Role requested: worker
[INFO] CA strict verification is set to true
[INFO] Using default agent configuration directory /etc/rancher/agent
[INFO] Using default agent var directory /var/lib/rancher/agent
[INFO] Successfully downloaded CA certificate
[INFO] Value from https://192.168.1.177.nip.io/cacerts is an x509 certificate
[INFO] Successfully tested Rancher connection
[INFO] Downloading rancher-system-agent binary from https://192.168.1.177.nip.io/assets/rancher-system-agent-amd64
[INFO] Successfully downloaded the rancher-system-agent binary.
[INFO] Downloading rancher-system-agent-uninstall.sh script from https://192.168.1.177.nip.io/assets/system-agent-uninstall.sh
[INFO] Successfully downloaded the rancher-system-agent-uninstall.sh script.
[INFO] Generating Cattle ID
[INFO] Cattle ID was already detected as d0b7a0944ed79cd66c187b81410ed303306033a2fc923876cb513025a4bfb76. Not generating a new one.
[INFO] Successfully downloaded Rancher connection information
[INFO] systemd: Creating service file
[INFO] Creating environment file /etc/systemd/system/rancher-system-agent.env
[INFO] Enabling rancher-system-agent.service
Created symlink /etc/systemd/system/multi-user.target.wants/rancher-system-agent.service → /etc/systemd/system/rancher-system-agent.service.
[INFO] Starting/restarting rancher-system-agent.service

```

이 후 **Provisioning Log**에서 설치 과정 확인이 가능합니다.

Cluster: custom-test
Updating
Explore
Show Configuration
⋮

Namespace
● fleet-default
Age
2 secs
Provisioner
RKE2

Labels and Annotations
0
There are no labels and annotations configured on this resource.
[Show Configuration](#)

Machines (1)
Provisioning Log
Registration
Snapshots (0)
Autoscaler (0)
Conditions (16)
Recent Events (0)
Related Resources (11)

1:07:20 pm [INFO] waiting for at least one control plane, etcd, and worker node to be registered
1:16:42 pm [INFO] waiting for viable init node
1:16:48 pm [INFO] configuring bootstrap node(s) custom-1e68cc7c4155: waiting for agent to check in and apply initial plan
1:17:04 pm [INFO] configuring bootstrap node(s) custom-1e68cc7c4155: waiting for probes: calico, etcd, kube-apiserver, kube-controller-manager, kube-scheduler, kubelet
1:18:14 pm [INFO] configuring bootstrap node(s) custom-1e68cc7c4155: waiting for probes: calico, etcd, kube-apiserver, kube-controller-manager, kube-scheduler
1:18:50 pm [INFO] configuring bootstrap node(s) custom-1e68cc7c4155: waiting for probes: calico, kube-apiserver, kube-controller-manager, kube-scheduler
1:19:14 pm [INFO] configuring bootstrap node(s) custom-1e68cc7c4155: waiting for probes: calico

해당 Log에서 provisioning done 및 상단의 상태가 **Active** 로 변경되면 Downstream Cluster 생성이 완료됩니다.

Cluster: custom-test Active Explore Show Configuration

Namespace: fleet-default Labels and Annotations: 0
 Age: 2 secs There are no labels and annotations configured on this resource.
 Provisioner: RKE2 [Show Configuration](#)

Machines (1) Provisioning Log Registration Snapshots (0) Autoscaler (0) Conditions (22) Recent Events (0) Related Resources (13)

```

1:07:20 pm [INFO] waiting for at least one control plane, etcd, and worker node to be registered
1:16:42 pm [INFO] waiting for viable init node
1:16:48 pm [INFO] configuring bootstrap node(s) custom-1e68cc7c4155: waiting for agent to check in and apply initial plan
1:17:04 pm [INFO] configuring bootstrap node(s) custom-1e68cc7c4155: waiting for probes: calico, etcd, kube-apiserver, kube-controller-manager, kube-scheduler, kubelet
1:18:14 pm [INFO] configuring bootstrap node(s) custom-1e68cc7c4155: waiting for probes: calico, etcd, kube-apiserver, kube-controller-manager, kube-scheduler
1:18:50 pm [INFO] configuring bootstrap node(s) custom-1e68cc7c4155: waiting for probes: calico, kube-apiserver, kube-controller-manager, kube-scheduler
1:19:14 pm [INFO] configuring bootstrap node(s) custom-1e68cc7c4155: waiting for probes: calico
1:21:16 pm [INFO] configuring bootstrap node(s) custom-1e68cc7c4155: waiting for cluster agent to connect
1:22:50 pm [INFO] non-ready bootstrap machine(s) custom-1e68cc7c4155 and join url to be available on bootstrap node
1:23:08 pm [INFO] configuring bootstrap node(s) custom-1e68cc7c4155: waiting for probes: etcd, kube-apiserver
1:23:10 pm [INFO] non-ready bootstrap machine(s) custom-1e68cc7c4155 and join url to be available on bootstrap node
1:23:12 pm [INFO] configuring bootstrap node(s) custom-1e68cc7c4155: waiting for probes: kube-controller-manager, kube-scheduler
1:23:16 pm [INFO] configuring bootstrap node(s) custom-1e68cc7c4155: waiting for plan to be applied
1:23:38 pm [INFO] custom-1e68cc7c4155
1:23:40 pm [INFO] non-ready bootstrap machine(s) custom-1e68cc7c4155 and join url to be available on bootstrap node
1:23:54 pm [INFO] provisioning done
  
```

2-2. SUSE Virtualization Cluster Create(시연)

Main > Cluster Management 메뉴에 접속 후 Create 버튼을 클릭합니다.

SUSE Rancher Prime

Home local test

GLOBAL APPS

- Cluster Management
- Continuous Delivery
- Virtualization Management

Clusters

Import Existing Create

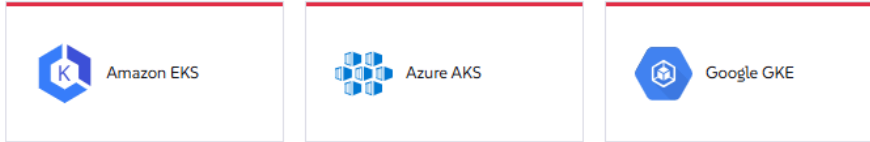
[Download KubeConfig](#) [Take Snapshot](#) [Download YAML](#) [Delete](#)

☐	State	Name	Version Architecture	Provider Distro	Autoscaler	Machines	Age	
☐	Active	local	v1.34.5+rke2r1 amd64	Local RKE2	—	3	25 days	Explore
☐	Active	test	v1.34.5+rke2r1 amd64	Harvester RKE2	—	5	13 hours	Explore

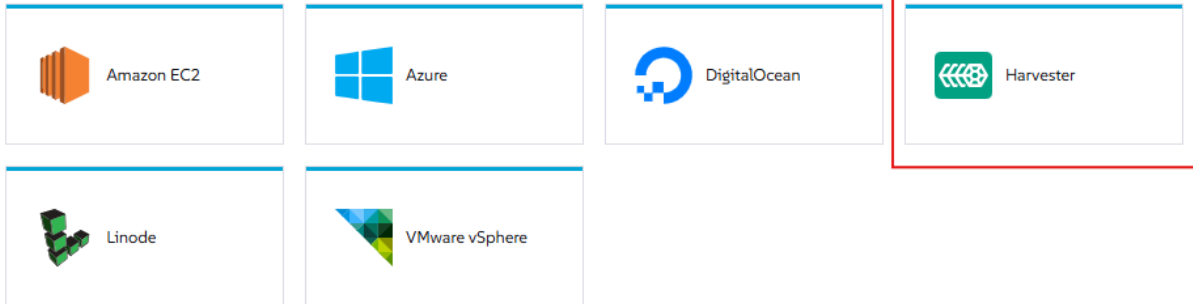
Harvester를 선택합니다.

Cluster: Create

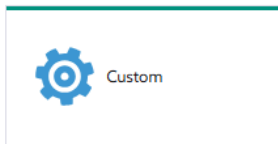
Create a cluster in a hosted Kubernetes provider



Provision new nodes and create a cluster using RKE2/K3s



Use existing nodes and create a cluster using RKE2/K3s



처음 SUSE Virtualization (구, Harvester) Provisioning을 진행할 때는 Cloud Credential을 생성해야 합니다. (Credential Name 및 Harvester Cluster 선택 후 Continue 클릭)

Cluster: Create Harvester

Ok, Let's create a new credential

Credential Name *
suse-virt-cred

Imported Harvester Cluster *
suse-virt

Select Existing

Cancel

Continue

Cluster Name 및 **SUSE Virtualization**를 통해 생성할 VM의 정보를 입력합니다.

- **Cluster Name:** `harvester-test`
- **SUSE Virtualization:** `v1.34.6+rke2r3`

Cluster: Create Harvester

Cloud Credential
suse-virt-cred

Cluster Name *
harvester-test

Cluster Appearance
HRT [Customize](#)

Cluster Description
Any text you want that better describes this cluster

Machine Pools

1 etcd
1 Control Plane
1 Worker

hav-cluster

Pool Name *
hav-cluster

Machine Count *
1

Roles
☒ etcd
☒ Control Plane
☒ Worker

CPUs *
4

Memory *
8

Namespace *
sr-edu99

SSH User *
e.g. ubuntu

Volumes

Image Volume

Image *
sles15-sp7-minimal-vm.x86_64-cloud-gm.qcow2 (cloud...

Disk *
102

bootOrder: 1

Add Volume
Add VM Image

Networks

Network

Network Name *

default/vm-network (vlanId=1)

Add Network

Hide Advanced

VGPIs

Add a new row

User Data:

User Data Template

common/sr-edu-user-template

```

1 #cloud-config
2 package_update: true
3 packages:
4   - qemu-guest-agent
5   - iptables
6 chpasswd:
7   list: |
8     root:SRdu!!!
9   expire: false
10  ssh_pwauth: true
11  runcmd:
12    - systemctl enable --now qemu-guest-agent.service
13    - sed -i 's/^\#PermitRootLogin,*/PermitRootLogin yes/' /etc/ssh/sshd_config
14    - systemctl restart sshd
15  - systemctl
16    - enable
17    - --now
18    - qemu-guest-agent.service
19  - SUSEConnect -r 2960AFA26DA69351

```

☒ Install guest agent
☐ Enable CPU Pinning
☐ Enable TPM
☐ Booting in EFI mode

종류	설명	구성
Pool Name	생성할 VM의 식별용 이름을 지정합니다.	hav-cluster
Machine Count	해당 풀에서 배포할 VM의 총 개수를 설정합니다.	1
Roles	해당 노드가 클러스터 내에서 수행할 Kubernetes 역할 (etcd, Control Plane, Worker)을 지정합니다. (모두 체크 시 단일 노드로 전체 역할 수행)	All Check
CPUs	각 가상 머신에 할당할 가상 CPU(vCPU) 코어 수를 설정합니다.	4
Memory	각 가상 머신에 할당할 메모리(RAM) 용량(GiB)을 설정합니다.	8
Namespace	SUSE-Virtualization 환경 내에서 가상 머신 리소스가 논리적으로 격리되어 생성될 네임스페이스를 지정합니다.	sr-edu-XX
SSH User	생성된 가상 머신에 SSH로 원격 접속할 때 사용할 기본 사용자 계정을 지정합니다.	root
Volumes	가상 머신 부팅에 사용할 OS 이미지를 선택하고 루트 디스크 볼륨 크기를 설정합니다.	sles15-sp7.qcow
Networks	가상 머신이 통신할 Harvester 내부 또는 외부 네트워크를 연결합니다.	vm-network

종류	설명	구성
User Data	Cloud-init을 통해 VM 초기 부팅 시 자동으로 적용할 설정(QEMU Guest Agent 설치, 패스워드 설정, SSH 허용, SUSEConnect 등록 등) 템플릿을 지정합니다.	sr-edu-user-template

생성할 Cluster의 정보를 입력 후 **Create** 버튼을 클릭합니다.
(해당 과정은 Custom Cluster 생성 과정과 동일합니다.)

Cluster Configuration

Basics

Member Roles

Add-on: Calico

Add-on: Harvester Cloud Provider

Additional Manifest

Agent Environment Vars

Cluster Agent

etcd

Fleet Agent

Labels & Annotations

Networking

Registries

Upgrade Strategy

Advanced

Basics

Kubernetes Version
v1.34.6+rke2r3

Cloud Provider
Harvester

☐ Show deprecated Kubernetes patch versions

Container Network
calico

Security

Pod Security Admission Configuration Template
Default - RKE2 Embedded

☐ Project Network Isolation

System Services

☒ CoreDNS ☒ NGINX Ingress ☒ Metrics Server

Cancel

Edit as YAML

Create

이 후 생성된 VM에 접속하여 아래 명령어를 통해 **iptables** 를 설치합니다.

```
zypper install -y iptables
```

Virtual Machines

Create

State	Name	CPU	Memory	IP Address	Node	Age
Namespace: sr-edu99						
☐ Running	harvester-test-hav-cluster-gn8d4-cs7g7	4	8 Gi		suse-virt02	Just now
☐ Running	sr-import-99	4	8 Gi	192.168.2.92	suse-virt02	6 days
☐ Running	sr-downstream-99	4	8 Gi	192.168.2.93	suse-virt01	6 days
☐ Running	sr-gui-99	2	4 Gi	192.168.2.91	suse-virt02	6 days
☐ Running	sr-rancher-99	4	8 Gi	192.168.2.90	suse-virt02	6 days

```

harvester-test-hav-cluster-gn8d4-cs7g7:~ # zypper install iptables
Refreshing service 'Basesystem_Module_15_SP7_x86_64'.
Refreshing service 'Python_3_Module_15_SP7_x86_64'.
Refreshing service 'SUSE_Linux_Enterprise_Server_15_SP7_x86_64'.
Refreshing service 'Server_Applications_Module_15_SP7_x86_64'.
Refreshing service 'Systems_Management_Module_15_SP7_x86_64'.
Loading repository data...
Reading installed packages...
Resolving package dependencies...

The following 6 NEW packages are going to be installed:
  iptables libip6tc2 libnetfilter_conntrack3 libnfnetlink0 libnfnetlink1 xtables-plugins
6 new packages to install.

Package download size: 620.6 KiB

Package install size change:
      1      2.3 MiB required by packages that will be installed
2.3 MiB 1 -      0 B released by packages that will be removed

Backend: classic_rpmtrans
Continue? [y/n/v/?...? shows all options] (y): y
  
```

이 후 Custom Cluster Create 과정과 동일하게 Provisioning Log 에서 생성 과정 확인을 할 수 있습니다.

Cluster: harvester-test
Updating
Explore
Show Configuration

Namespace
fleet-default
Age
3.4 mins
Provisioner
RKE2
Machine Provider
Harvester
Kubernetes Version
v1.34.6+rke2r3
Machine Pools
1
Machines
1

Labels and Annotations
0
There are no labels and annotations configured on this resource.
[Show Configuration](#)

Machine Pools (1)
Provisioning Log
Snapshots (0)
Autoscaler (0)
Conditions (17)
Recent Events (0)
Related Resources (12)

```

2:39:02 pm [INFO ] waiting for at least one control plane, etcd, and worker node to be registered
2:39:04 pm [INFO ] waiting for viable init node
2:40:40 pm [INFO ] configuring bootstrap node(s) harvester-test-hav-cluster-gn8d4-cs7g7: waiting for agent to check in and apply initial plan
2:41:24 pm [INFO ] configuring bootstrap node(s) harvester-test-hav-cluster-gn8d4-cs7g7: waiting for probes: calico, etcd, kube-apiserver, kube-
controller-manager, kube-scheduler, kubelet
2:41:54 pm [INFO ] configuring bootstrap node(s) harvester-test-hav-cluster-gn8d4-cs7g7: waiting for probes: calico, etcd, kube-apiserver, kube-
controller-manager, kube-scheduler
2:42:14 pm [INFO ] configuring bootstrap node(s) harvester-test-hav-cluster-gn8d4-cs7g7: waiting for probes: calico, kube-apiserver, kube-
controller-manager, kube-scheduler
2:42:20 pm [INFO ] configuring bootstrap node(s) harvester-test-hav-cluster-gn8d4-cs7g7: waiting for probes: calico, kube-controller-manager,
kube-scheduler
2:42:30 pm [INFO ] configuring bootstrap node(s) harvester-test-hav-cluster-gn8d4-cs7g7: waiting for probes: calico

```

해당 Log에서 **provisioning done** 및 상단의 상태가 **Active** 로 변경되면 Downstream Cluster 생성이 완료됩니다.

Cluster: harvester-test
Active
Explore
Show Configuration

Namespace
fleet-default
Age
6 mins
Provisioner
RKE2
Machine Provider
Harvester
Kubernetes Version
v1.34.6+rke2r3
Machine Pools
1
Machines
1

Labels and Annotations
0
There are no labels and annotations configured on this resource.
[Show Configuration](#)

Machine Pools (1)
Provisioning Log
Snapshots (0)
Autoscaler (0)
Conditions (23)
Recent Events (0)
Related Resources (14)

```

2:39:02 pm [INFO ] waiting for at least one control plane, etcd, and worker node to be registered
2:39:04 pm [INFO ] waiting for viable init node
2:40:40 pm [INFO ] configuring bootstrap node(s) harvester-test-hav-cluster-gn8d4-cs7g7: waiting for agent to check in and apply initial plan
2:41:24 pm [INFO ] configuring bootstrap node(s) harvester-test-hav-cluster-gn8d4-cs7g7: waiting for probes: calico, etcd, kube-apiserver, kube-
controller-manager, kube-scheduler, kubelet
2:41:54 pm [INFO ] configuring bootstrap node(s) harvester-test-hav-cluster-gn8d4-cs7g7: waiting for probes: calico, etcd, kube-apiserver, kube-
controller-manager, kube-scheduler
2:42:14 pm [INFO ] configuring bootstrap node(s) harvester-test-hav-cluster-gn8d4-cs7g7: waiting for probes: calico, kube-apiserver, kube-
controller-manager, kube-scheduler
2:42:20 pm [INFO ] configuring bootstrap node(s) harvester-test-hav-cluster-gn8d4-cs7g7: waiting for probes: calico, kube-controller-manager,
kube-scheduler
2:42:30 pm [INFO ] configuring bootstrap node(s) harvester-test-hav-cluster-gn8d4-cs7g7: waiting for probes: calico
2:43:58 pm [INFO ] configuring bootstrap node(s) harvester-test-hav-cluster-gn8d4-cs7g7: waiting for cluster agent to connect
2:45:34 pm [INFO ] non-ready bootstrap machine(s) harvester-test-hav-cluster-gn8d4-cs7g7 and join url to be available on bootstrap node
2:45:38 pm [INFO ] configuring bootstrap node(s) harvester-test-hav-cluster-gn8d4-cs7g7: waiting for plan to be applied
2:45:46 pm [INFO ] non-ready bootstrap machine(s) harvester-test-hav-cluster-gn8d4-cs7g7 and join url to be available on bootstrap node
2:46:02 pm [INFO ] provisioning done

```

Step 3. Existing Cluster Import 수행(시연)

Cluster Import 방식은 미리 생성된 RKE2 Cluster가 필요합니다.

(아래 VM에는 RKE2 Kubernetes 구성을 완료한 VM입니다. 해당 VM의 IP를 저장합니다.)

Virtual Machines

[Create](#)

[Stop](#) [Restart](#) [Download YAML](#) [Delete](#)

[Filter](#)

☐	State	Name	CPU	Memory	IP Address	Node	Age
Namespace: sr-edu99							
☐	Running	harvester-test-hav-cluster-gn8d4-cs7g7	4	8 Gi	192.168.2.104	suse-virt02	7 mins
☐	Running	sr-import-99	4	8 Gi	192.168.2.92	suse-virt02	6 days
☐	Running	sr-downstream-99	4	8 Gi	192.168.2.93	suse-virt01	6 days
☐	Running	sr-gui-99	2	4 Gi	192.168.2.91	suse-virt02	6 days
☐	Running	sr-rancher-99	4	8 Gi	192.168.2.90	suse-virt02	7 days

3-1. Generic Cluster Import

Main > Cluster Management 메뉴에 접속 후 [Import Existing](#) 버튼을 클릭합니다.

[Home](#)

[local](#)

[test](#)

[Cluster Management](#)

[Continuous Delivery](#)

[Virtualization Management](#)

[Download KubeConfig](#) [Take Snapshot](#) [Download YAML](#) [Delete](#)

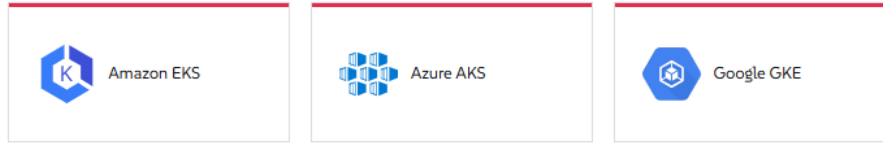
[Filter](#)

☐	State	Name	Version Architecture	Provider Distro	Autoscaler	Machines	Age	
☐	Active	local	v1.34.5+rke2r1 amd64	Local RKE2	—	3	25 days	Explore
☐	Active	test	v1.34.5+rke2r1 amd64	Harvester RKE2	—	5	13 hours	Explore

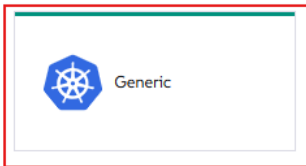
Generic을 선택합니다.

Cluster: Import

Register an existing cluster in a hosted Kubernetes provider



Import any Kubernetes cluster



Cluster Name을 입력하고 Create 버튼을 클릭합니다.

Cluster: Import Generic

Import Harvester Clusters via [Virtualization Management](#) ×

Cluster Name *
import-test

Cluster Description
Any text you want that better describes this cluster

Basics

K8s version management

Future changes to the system default setting will result in redeployment of the cluster agent.

☒ Global default (enabled)
☐ Enabled
☐ Disabled

Version management for imported clusters is globally enabled.

Member Roles

User	Role
☒ Default Admin (admin)	Cluster Owner
☐ Local	

Add

Cancel

Create

이후 아래와 같은 화면이 나타나면, Step 3에서 확인한 IP를 사용해 `import VM`에 SSH로 접속합니다.

접속 후 화면에 표시된 두 번째 명령어를 복사하여 터미널에서 실행합니다.

Cluster: import-test Provisioning Explore Show Configuration ⋮

Namespace fleet-default
Age 1secs
Provisioner Imported

Labels 0
There are no labels configured on this resource.
[Show Configuration](#)

Annotations 1
rancher.io/imported-cluster-version-management: system-...

Provisioning Log
Registration
Autoscaler (0)
Conditions (15)
Recent Events (0)
Related Resources (9)

You should not import a cluster which has already been connected to another instance of Rancher as it will lead to data corruption.

Run the `kubectl` command below on an existing Kubernetes cluster running a supported Kubernetes version to import it into Rancher Prime:

```
kubectl apply -f https://192.168.1.177.nip.io/v3/import/x69cwbft7qgzv8662rwp6vdxskfnlk7vb4p19945k6hncwvnx1q27_c-d72r2.yaml
```

If you get a "certificate signed by unknown authority" error, your Rancher Prime installation has a self-signed or untrusted SSL certificate. Run the command below instead to bypass the certificate verification:

```
curl --insecure -sfl https://192.168.1.177.nip.io/v3/import/x69cwbft7qgzv8662rwp6vdxskfnlk7vb4p19945k6hncwvnx1q27_c-d72r2.yaml | kubectl apply -f -
```

If you get permission errors creating some of the resources, your user may not have the `cluster-admin` role. Use this command to apply it:

```
kubectl create clusterrolebinding cluster-admin-binding --clusterrole cluster-admin --user <your username from your kubeconfig>
```

```
localhost:~ # ssh 192.168.2.92
The authenticity of host '192.168.2.92 (192.168.2.92)' can't be established.
ED25519 key fingerprint is SHA256:jmByQ2k0shnzf8PK09avHKgodzS00NCnR+Cy7DRaBL4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.2.92' (ED25519) to the list of known hosts.
(root@192.168.2.92) Password:
Last login: Fri Apr 17 06:08:21 2026
sr-import-99:~ # curl --insecure -sfl https://192.168.1.177.nip.io/v3/import/x69cwbft7qgzv8662rwp6vdxskfnlk7vb4p19945k6hncwvnx1q27_c-d72r2.yaml | kubectl app
ly -f -
clusterrole.rbac.authorization.k8s.io/proxy-clusterrole-kubeprierver created
clusterrolebinding.rbac.authorization.k8s.io/proxy-role-binding-kubernetes-master created
namespace/cattle-system created
serviceaccount/cattle created
clusterrolebinding.rbac.authorization.k8s.io/cattle-admin-binding created
secret/cattle-credentials-53d5c35b7d created
clusterrole.rbac.authorization.k8s.io/cattle-admin created
deployment.apps/cattle-cluster-agent created
service/cattle-cluster-agent created
sr-import-99:~ #
```

해당 Command 입력 후 Cluster의 상태와 Machine Pool의 State가 Active 상태가 되면 Import를 통한 Cluster 생성이 완료됩니다.

Cluster: import-test Active Explore Show Configuration ⋮

Namespace fleet-default
Age 1secs
Provisioner RKE2

Labels 1
provider.cattle.io: rke2

Annotations 1
rancher.io/imported-cluster-version-management: system-...

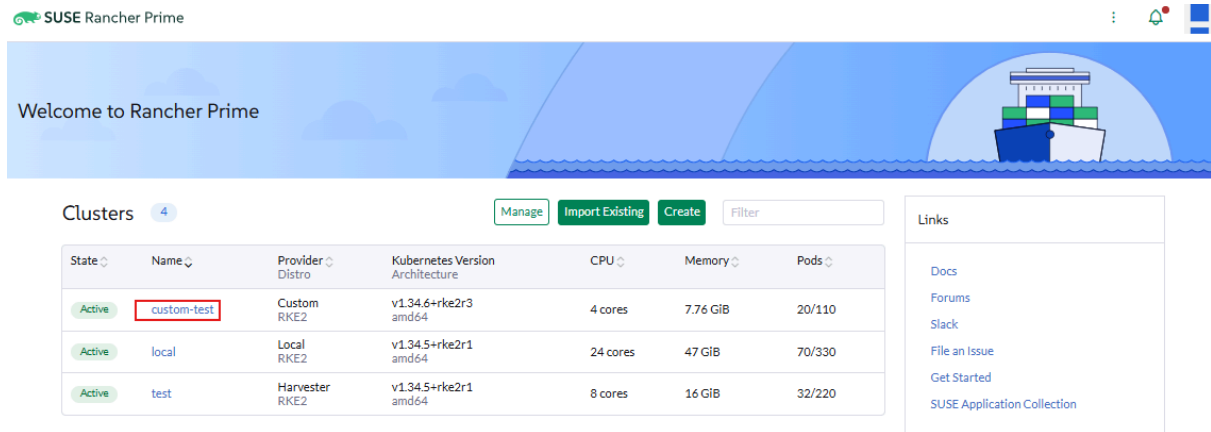
Machine Pools (1)
Provisioning Log
Autoscaler (0)
Conditions (20)
Recent Events (0)
Related Resources (11)

Download YAML
⋮
Filter

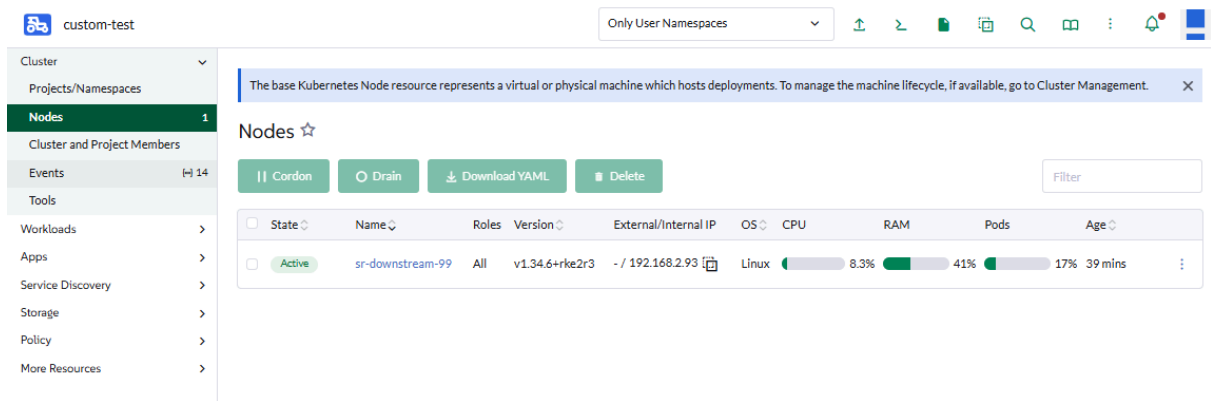
☐ State	Name	Node	External/Internal IP	OS	Roles	Age
Not in a Pool						
☐ Active	machine-4gqbk	sr-import-99	- / -	Linux	Control Plane, Etc	3.2 mins

Step 4. Cluster 및 노드 상태 확인

메인 대시보드에서 생성되어 있는 Cluster 선택



Cluster > Nodes 메뉴에서 노드 상태 확인이 가능합니다.



4. 성공 기준

#	기준	확인
1	Rancher Dashboard 확인	☐
2	Downstream Cluster 생성 방식 이해	☐
3	Downstream Cluster 생성 완료	☐

5. 실습 결과 요약

- **Rancher Dashboard 구조 숙지:** Rancher 메인 대시보드 및 클러스터 대시보드를 탐색하며, Workloads, Service Discovery, Storage 등 쿠버네티스 리소스 관리를 위한 주요 메뉴의 역할과 위치를 파악했습니다.

- **클러스터 연동 방식 이해:** Rancher 환경에서 새로운 쿠버네티스 클러스터를 직접 구축하는 Provision 방식과, 이미 구성된 외부 클러스터를 연결하는 Import 방식의 차이점을 명확히 이해했습니다.
 - **Downstream 클러스터 프로비저닝 완료:** 실제 실습 환경에서 Custom 클러스터를 직접 생성하여 노드에 적절한 역할을 할당하고, 정상적으로 Rancher에 등록되어 통합 관리할 수 있는 상태를 만들었습니다.
-

 **다음 세션으로 넘어가기**

 **메인 페이지로 돌아가기**